

Kerala Government Governed LLM, Ontology, Typed Analytics & Calibrated Graph Platform

**Corrected Architecture v5 - Tiered Assurance, Explicit Method Binding, Standard Data
Contracts,
Pre-Execution Governance, Honest Search-Coverage Claims, and Delivery Capacity**

Core architectural correction

v5 accepts that the hardest trust boundary is not SQL generation but interpretation: mapping a natural-language question to the right analytical method remains partly a judgment problem. The architecture therefore inserts a canonical Question Specification, deterministic contract routing where possible, ambiguity detection, explicit method-binding attestation, and human/user clarification when multiple methods remain plausible.

v5 also separates a strict Governed Decision-Support Lane from an Exploratory Research Lane. This resolves the tension between a government-wide analytical environment and a pre-registered-contract model: broad exploration is allowed under lower assurance in an isolated read-only sandbox, while consequential or reusable outputs must be promoted into governed contracts before they are treated as certified decision support.

**Design goal: maximize useful government-wide analysis while making assurance level,
unresolved judgment, organizational cost, and unknown search coverage explicit.**

Claim boundary

The platform does not guarantee that an LLM has understood the user's true intent, does not guarantee that all valuable question types can be pre-registered, and cannot generally measure real-world graph-search recall when ground truth is unknown. v5 converts those limits into explicit operating modes, stop states, attestations and project-capacity assumptions rather than hiding them behind mechanically valid contracts.

Document Map

Part	Sections	Focus
A	1-8	Mission, scope, two-lane model, five assurance dimensions, reference architecture and claim boundaries.
B	9-18	Canonical Question Specification, method binding, applicability, Method Contracts, knowledge-acquisition and governance throughput.
C	19-25	Standard data-engineering terminology, effective-dated reference data/data contracts, typed frames, conformed interfaces and STOP governance.
D	26-34	Graph search, exposure governance, honest coverage evaluation, evidence separation and adaptive search.
E	35-40	Pre-execution gate, plan compilation, execution ordering, methodological/evidentiary gates and audit.
F	41-47	Operating model, staffing, review capacity, timeline, MVP, success metrics and cost/capacity controls.
G	48-52	Worked examples, residual risks, non-negotiable rules and conclusion.
Appendices	A-I	Question/method schemas, data-contract schema, STOP taxonomy, discovery coverage certificate, staffing assumptions, v4 migration, and central government data platform/database design.

1. Executive Summary

The platform is a government-wide analytical substrate with tiered assurance, not a promise that every government question can be answered through a pre-certified path. A high-end LLM may interpret questions, propose analysis designs and orchestrate tools, but the architecture no longer treats successful binding to an internally consistent Method Contract as evidence that the method was the right interpretation of the user's question.

v5 inserts an explicit interpretive boundary between natural language and governed execution. The LLM first produces a Canonical Question Specification containing population, outcome/exposure roles, requested comparison, time basis, decision context and consequence class. A deterministic Method Router uses those declared fields to enumerate eligible Method Contracts. When exactly one registered method matches and its applicability conditions pass, execution can proceed. When multiple methods remain plausible, the system must ask for clarification or obtain an authorized method-binding attestation; it may not silently choose one.

The second correction is operational. Pre-registration is intentionally not required for all exploratory work. The Governed Decision-Support Lane remains contract-gated and reproducible. A separate Exploratory Research Lane permits broader read-only analysis over approved semantic views inside an isolated sandbox, with strong labeling that results are exploratory and cannot directly support consequential action. Recurring or high-value exploratory patterns are promoted into governed Method Contracts, bridges or primitives.

The third correction is epistemic. Search exposure remains governed, but validation no longer pretends that seeded fixtures provide real-world recall. When true labels are absent, the Search Coverage Certificate reports true candidate recall as NOT_ESTIMABLE and separately reports seeded-fixture detection, larger-budget reference recovery, exploration yield, adversarial red-team results and exposure diagnostics.

The fourth correction is practical. Effective-dated administrative mappings are described using standard data-engineering concepts - Type-2 history/effective dating, master/reference-data management, data contracts, quality assertions and lineage/impact propagation - rather than being presented as a novel Kerala-specific mechanism. Finally, v5 adds a staffing envelope, review-capacity assumptions, target governance lead times and a staged 15-month pilot/expansion plan so the document functions as a project plan rather than only an architecture specification.

2. What v5 Corrects

Issue	Why the v4 posture was insufficient	v5 correction
Method selection remains a trust gap	A Method Contract engine can validate internal conformance but cannot prove that the LLM selected the right contract for a natural-language question.	Insert Canonical Question Specification, deterministic eligible-contract routing, ambiguity detection, binding evidence, explicit clarification/attestation and a new Interpretive/Binding assurance dimension.
Knowledge-acquisition bottleneck	Every new governed question class still requires steward-authored methods/bridges; government-wide long-tail demand would create STOP or approval queues.	Introduce two lanes: broad exploratory sandbox for non-consequential research; governed lane for reusable/consequential analysis. Use method families/templates and promote recurring exploration patterns.

Issue	Why the v4 posture was insufficient	v5 correction
Search recall is circular without ground truth	Seeded fixtures measure detection of known constructed patterns, not unknown real adversarial candidates.	Never report real-world candidate recall unless a defensible labeled universe exists. Use separate surrogate diagnostics and label true recall NOT_ESTIMABLE.
Standard data engineering was renamed	Effective-dated mappings, health checks and impact propagation mostly reuse established SCD/history, MDM/reference-data, data-contract and lineage patterns.	Adopt standard terminology and explicitly state that novelty lies in integration/governance boundaries, not the underlying data-engineering mechanisms.
Execution ordering	Graph search could consume a large resource/exposure budget before the Computational Gate approved authorization and resource limits.	Split governance into a Pre-Execution Authorization & Resource Gate and a post-planning Computational Gate. No expensive graph/analytics execution occurs before preflight passes.
No delivery-capacity model	The roadmap assumed multiple steward roles and review authorities without staffing, queue or schedule assumptions.	Add staffing ranges, review-capacity targets, SLAs, delivery milestones, load-shedding rules and coverage metrics.
Mission tension	Government-wide exploration and pre-registered contract-only execution were asserted simultaneously.	Define government-wide as substrate/access breadth with tiered assurance, not universal governed answerability.

3. Mission and Scope

Mission statement

Build a government-wide analytical substrate that supports broad, purpose-authorized exploration and a narrower, high-assurance governed path for reusable or consequential decision support. The platform must make the assurance level visible at every step and must not collapse exploratory analysis, governed method execution and authoritative decision-making into one trust category.

- Government-wide means common identity, semantic, reference-data, analytical and graph infrastructure can be reused across departments where policy permits.
- It does not mean every natural-language question has a registered Method Contract or certified cross-domain bridge.
- The primary success metric is not "percentage of questions answered somehow"; it is the distribution of questions across Governed PASS, Governed with limitations, Exploratory-only, Human-review required, and STOP states.
- Consequential use always requires the governed lane or an explicitly authorized exceptional process.

4. Two-Lane Operating Model

Lane	Purpose	Allowed execution	Output status	Promotion path
Governed Decision-Support	Reusable, routine or consequential analyses where method and semantics must be reproducible.	Registered primitives, typed frames, active data/reference contracts, registered bridges, Method Contracts, governed graph searches.	GOVERNED_PASS / PASS_WITH_LIMITATIONS / HUMAN_APPROVAL_REQUIRED / STOP.	Versioned release and change control.
Exploratory Research	Long-tail question discovery, hypothesis generation, analyst investigation and contract prototyping.	Read-only isolated sandbox over approved semantic views; bounded generated SQL or notebooks may be allowed subject to purpose, minimization, row/resource limits and audit.	EXPLORATORY_NOT_CERTIFIED. Cannot directly justify adverse or consequential action.	Promote recurring/high-value workflows into governed primitive/bridge/method contracts after tests and steward review.

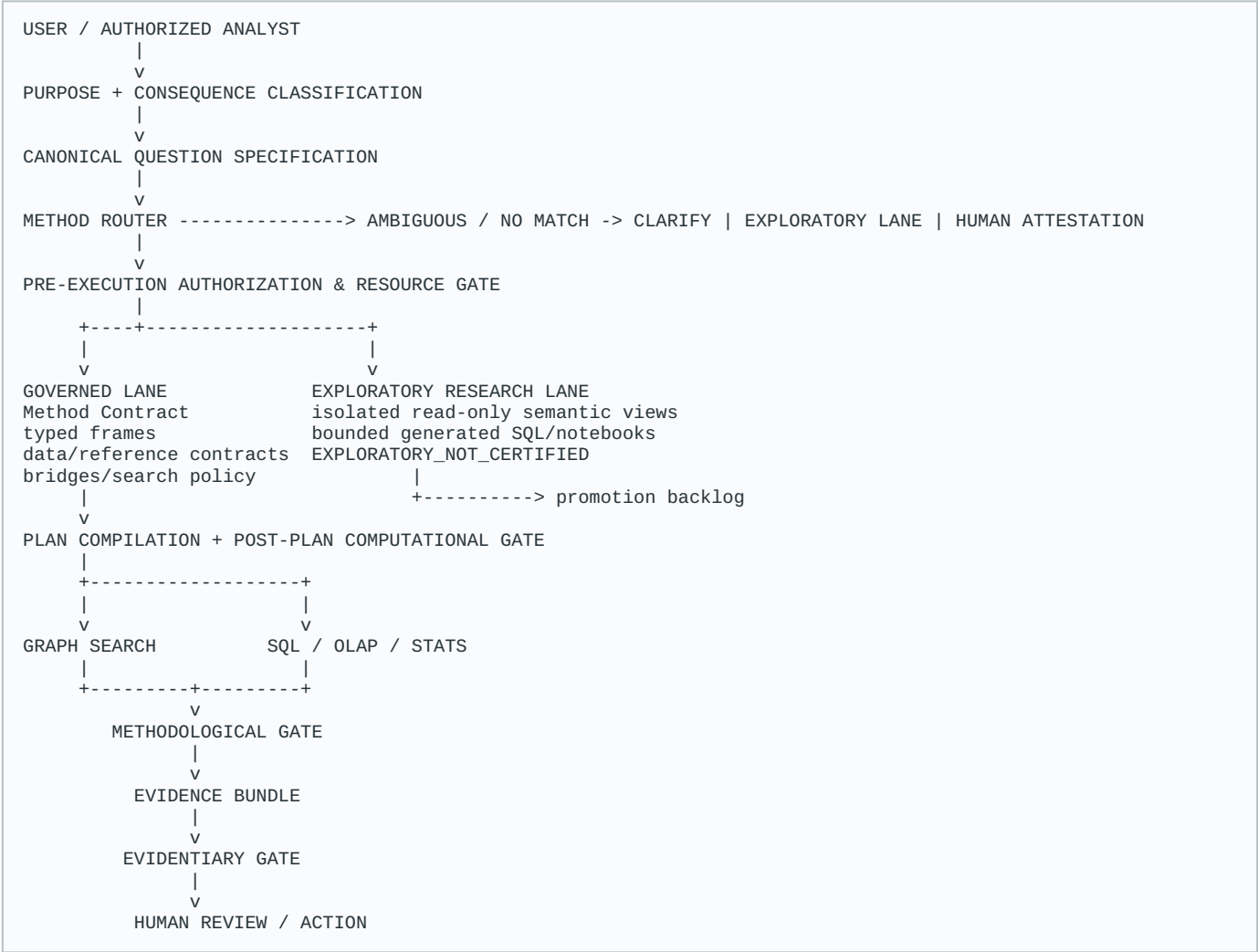
Critical separation

No exploratory result becomes "certified" merely because the SQL ran successfully. Promotion requires semantic review, method applicability tests, data/reference contracts, regression fixtures and the appropriate governance tier.

5. Five Assurance Dimensions

Assurance dimension	Question	Primary mechanisms	What it cannot prove alone
Interpretive / Method-binding validity	Did the selected method actually correspond to the user's canonicalized question and decision context?	Question Specification, deterministic router, ambiguity detection, applicability tests, clarification/attestation.	The user's latent true intent or every domain nuance.
Computational composition safety	Was a registered plan executed consistently with declared grain, measure, time and data/reference contracts?	Typed frames, operator signatures, active data contracts, computational gate.	That the chosen method/comparison is appropriate.
Methodological validity	Does the bound method use defensible peers, denominators, baselines, diagnostics and robustness checks?	Method Contract, applicability tests, robustness suite, expert escalation.	Absolute truth, all confounders, or universal causal validity.
Search-exposure integrity	Did finite-budget search systematically hide candidate classes before evidence evaluation?	Search policy audit, exploration budget, reference runs, suppression diagnostics, drift tests.	True recall when the qualifying universe is unknown.
Evidentiary sufficiency	Does the assembled evidence justify the stated claim?	Evidence models, evidence ladder, counter-evidence, human review.	Authority beyond the legal/administrative mandate.

6. Reference Architecture v5



7. Trust Boundaries

Component	May propose?	What it owns	Explicit limitation
LLM analyst	Yes	Natural-language interpretation, Question Specification draft, candidate method proposals, orchestration, explanation.	Cannot self-certify that its chosen method matches the user's intent.
Method Router	No	Deterministic eligibility filtering over registered semantic roles and method applicability rules.	Cannot resolve genuine ambiguity when several methods remain valid.
Authorized analyst / method approver	Yes	Binding attestation for ambiguous, novel or high-consequence designs.	Human judgment remains fallible and capacity-limited.
Typed composition engine	No	Mechanical operator/frame compatibility.	No methodological guarantee.
Data/reference contract service	No	Effective dating, validity assertions, quality health, versioning and impact propagation.	Cannot decide whether an administrative mapping is the substantively right attribution basis for a question.
Search manager	Policy-bounded	Candidate discovery within declared graph/resource envelope.	Finite-budget visibility is incomplete.
Evidence models	Model-specific	Domain-calibrated evidence or anomaly estimation where supported.	No universal probability of wrongdoing.
Authorized review process	Yes	Consequential interpretation and action within mandate.	Subject to legal and administrative process.

8. Honest Claim Boundaries

- v5 does not solve the semantic interpretation problem; it makes the handoff from interpretation to governed execution explicit and auditable.
- v5 does not remove the knowledge-acquisition bottleneck; it reduces its effect by separating exploratory breadth from governed assurance and by promoting reusable patterns rather than pre-registering the entire long tail.
- v5 does not claim real-world graph-search recall where ground truth is absent.
- v5 does not claim novelty for SCD/effective dating, data contracts, reference-data management, quality assertions or lineage. These are established engineering mechanisms used as building blocks.
- v5 does not promise that the governed lane answers every cross-domain question. STOP remains a legitimate outcome.

9. Canonical Question Specification

Before method selection, the platform converts natural language into an explicit question object that can be reviewed, clarified and versioned.

```
question_spec:
  original_question: "Are hospitals receiving increased funding without output improvement?"
  purpose: operational_resource_review
  consequence_class: medium
  population:
    entity_type: Hospital
    inclusion_rule: government_general_hospitals
  exposure:
    semantic_role: funding
    metric_family: operating_funding
  outcomes:
    - surgeries
    - admissions
    - outpatient_visits
  comparison_request:
    form: change_relative_to_peers
  time:
    basis: fiscal_year
    period: FY2022-23..FY2025-26
  causal_language_requested: false
  attribution_basis:
    unresolved: true
  ambiguity_flags:
    - "peer population not specified"
    - "operating vs total funding inferred from context"
```

Rule

Method selection is based on the canonical Question Specification, not directly on the raw prompt. Material inferred fields must either be confirmed by the user/authorized analyst or be covered by an explicit default rule whose use is disclosed.

10. Method Binding and Deterministic Routing

The LLM may propose methods, but it does not own the final eligibility decision. A deterministic router filters the Method Registry using declared question roles and contract applicability.

Routing step	Behavior
1. Resolve semantic roles	Map exposure, outcome, population, time basis and requested comparison into registered semantic categories.
2. Enumerate eligible contracts	Filter active contracts whose question class, entity types, outcome/exposure roles and consequence class are compatible.
3. Run applicability predicates	Check required data coverage, minimum support, attribution basis, temporal span and policy conditions.
4. Detect ambiguity	If zero contracts: MISSING_METHOD. If more than one materially different contract remains: METHOD_BINDING_AMBIGUOUS.
5. Bind or escalate	Exactly one eligible contract may bind automatically for routine low/medium consequence. Ambiguous/high-consequence cases require clarification or authorized attestation.

11. Method-Binding Statuses

Status	Meaning	Allowed next step
BINDING_DETERMINISTIC	One active contract uniquely matches the confirmed Question Specification and applicability rules.	Proceed to pre-execution gate.
BINDING_DEFAULTED	One match after a registered default was applied to a non-material field.	Proceed only with disclosed default; user/analyst may override.
METHOD_BINDING_AMBIGUOUS	Two or more materially different methods remain plausible.	Ask clarification or obtain authorized method-binding attestation.

Status	Meaning	Allowed next step
MISSING_METHOD	No registered governed method supports the specification.	Use exploratory lane, create backlog item, or stop.
METHOD_OUT_OF_SCOPE	A contract exists but applicability tests fail for this population/ data/context.	Stop or redesign; do not force binding.
HUMAN_METHOD_ATTESTATION	Authorized analyst explicitly selects/approves a method for a novel/high-consequence specification.	Proceed with approver identity, rationale and limits recorded.

12. Method Contracts

Contract element	Machine-checkable portion	Residual judgment
Question class / semantic roles	Enumerated roles and required fields.	Whether the real-world question has been correctly represented in those fields.
Peer template / baseline	Allowed templates, bounded parameters, minimum support.	Whether the template captures the most relevant contextual comparators.
Estimand / comparison form	Allowed estimands and claim language.	Whether a different estimand would better serve the decision context.
Denominator / eligibility version	Compatibility with metric and scheme versions.	Whether denominator construction reflects policy reality.
Diagnostics / robustness	Required checks and thresholds.	How to interpret borderline or conflicting diagnostics.
Interpretation limits	Permitted descriptive/associational/causal language.	Whether unmeasured context invalidates the practical inference.

13. Method Applicability Tests

- Population applicability: entity class, geography, scheme/facility type and inclusion/exclusion boundaries.
- Data sufficiency: reporting coverage, missingness, version consistency, minimum sample/peer support.
- Temporal applicability: period length, partial-year handling, policy/metric regime changes.
- Attribution applicability: administrative location vs service catchment vs funding-accountability basis.
- Consequence applicability: whether the method may be used for descriptive planning only or for higher-consequence review.
- Out-of-distribution flags: new institution types, exceptional events, restructurings or policy shocks outside the contract's regression corpus.

14. Novelty and High-Consequence Classification

Novelty may not be decided solely by the LLM. Rule-based triggers and registry comparisons define minimum escalation conditions.

Trigger	Required action
No eligible Method Contract	Exploratory lane or method design review; no governed auto-execution.
Multiple materially different eligible contracts	Clarification or method-binding attestation.
New peer predicate / new attribution basis / new estimand	Method steward review before governed use.
Adverse eligibility, enforcement, policing, tax penalty, blacklisting or similar consequence	Mandatory authorized pre-execution review regardless of apparent contract match.
Material data-contract degradation or scheme/reorganization break	Pause or downgrade until contract impact resolved.
Bridge declares material lossiness relevant to the method	Human review or alternative attribution.

15. Methodological Gate v5

- Verify Method Binding status and evidence; the gate may reject a contract that was syntactically valid but selected through unresolved ambiguity.
- Validate population, peer, baseline, estimand, denominator, attribution and exclusion bindings.

- Run required diagnostics and robustness checks.
- Check applicability/OOD flags and data/reference contract health.
- Produce PASS, PASS_WITH_LIMITATIONS, HUMAN_APPROVAL_REQUIRED, REJECTED_METHOD or METHOD_BINDING_INVALID.
- Record which checks were mechanical and which relied on human attestation.

16. Knowledge-Acquisition Strategy

The platform treats contract authoring as a scarce governance resource and designs for reuse rather than assuming unlimited steward capacity.

Mechanism	Purpose
Method families and inheritance	Factor common requirements into reusable method families, with domain-specific overrides rather than full duplicate contracts.
Conformed semantic roles	Reuse common roles such as exposure, outcome, population, period, denominator and peer template across domains.
Draft generation	LLM/tooling may generate a proposed contract from an exploratory workflow, but it remains DRAFT until tests and steward approval.
Regression corpus reuse	Each approved contract carries example questions, known counterexamples, rejected bindings and synthetic edge cases.
Promotion by observed demand	Create governed contracts when exploratory usage, consequence level or recurrence justifies the maintenance cost.
Deprecation	Low-use/obsolete contracts are retired rather than accumulating indefinitely.

17. Promotion from Exploratory to Governed

1. Capture the exploratory question, query plan, data sources, analyst interpretation and limitations.
2. Cluster repeated question patterns and estimate expected reuse/consequence.
3. Draft or extend semantic primitives/bridges/method family.
4. Create applicability tests, regression cases and known-counterexample tests.
5. Review data/reference contracts, policy boundaries and privacy implications.
6. Approve and version the governed contract.
7. Re-run representative historical exploratory cases through the governed path and compare outputs before promotion.

18. Governance Tiers and Capacity Targets

Tier	Change	Target review posture*	Capacity control
0	Existing governed contract and bounded parameters.	Immediate automated execution.	No steward queue.
1	New instance within approved template; low-consequence.	Target 3-5 business days where human sign-off is required.	Batch approvals; automated regression.
2	New reusable bridge/method family or attribution basis.	Target 2-4 weeks.	Prioritize by reuse/consequence; WIP limits.
3	Bespoke cross-domain semantics, novel high-consequence method, policy-sensitive linkage.	Target 4-8+ weeks; may require legal/policy review.	Explicit project intake and sponsor; not part of routine queue.

*Planning targets, not guaranteed service commitments. Actual lead time depends on staffing, legal/policy review and data readiness.

19. Standard Terminology and What Is Actually Novel

v4 term / mechanism	Standard engineering concept used in v5	v5 posture
Temporal Dependency Registry	Effective-dated reference/master data; Type-2 history pattern; data contracts; quality assertions; lineage/impact analysis.	Use standard terms in implementation and hiring. Do not claim novelty.
Dependency health states	Data-contract/data-quality status and release gating.	Retain because operationally useful, but map to familiar tooling concepts.
Dependency impact graph	Data lineage / downstream impact analysis.	Implement using standard metadata/catalog/lineage capabilities where possible.
Conformed interfaces	Conformed dimensions/metrics and semantic-layer contracts.	Standard concept adapted to typed analytical frames.
Method Contracts	Versioned analytical protocol / method specification plus executable checks.	Potentially distinctive in integration with LLM routing, but still grounded in established statistical governance practice.
Search exposure governance	Retrieval/ranking system evaluation under finite graph-search budgets.	Architectural integration is important; do not confuse with evidence-model calibration.

Novelty statement

The architectural contribution is the integrated trust model and assurance boundaries across interpretation, analytical contracts, typed composition, graph retrieval, evidence and human decision governance. The underlying reference-data history, data-contract, lineage and quality mechanisms should be implemented using established patterns and products wherever practical.

20. Effective-Dated Reference Data

<pre>reference_assignment: entity: Hospital_123 attribute: district value: District_07 effective_from: 2024-04-01 effective_to: 2025-09-30 source_system: health_facility_registry source_record_version: 981223 assignment_basis: administrative_location load_timestamp: 2025-10-01T02:15:00Z</pre>

- Use effective dates for business validity and separate load/audit timestamps for system history.
- Preserve prior versions rather than overwriting administrative history.
- Treat alternative attribution bases - administrative location, service catchment, accounting responsibility - as separate governed mappings, not one universal dependency.
- Historical analyses must pin the mapping basis and version appropriate to the question.

21. Data and Reference Contracts

Contract check	Examples
Schema and key expectations	Required identifiers present; uniqueness or expected multiplicity; valid date ranges.
Referential integrity	Hospital assignment resolves to a valid district version for the requested period.
Cardinality assertions	A hospital has at most one active administrative district at a point in time unless explicitly modeled otherwise.
Freshness/completeness	Registry watermark within SLA; missing assignment rate below threshold.
Semantic version compatibility	Metric, scheme and reference-data versions compatible with requested method.
Change-event handling	Reorganization triggers downstream impact assessment and contract revalidation.

22. Contract Health, Lineage and Historical Replay

State	Meaning	Execution policy
ACTIVE	Assertions pass for intended coverage.	Governed use permitted.
DEGRADED	Non-critical quality/freshness issues or partial coverage.	Only methods explicitly tolerant of the limitation may run; limitation must surface.
QUARANTINED	Material contract breach or unresolved mapping conflict.	No new governed execution depending on the affected mapping.
RETIRED	Superseded version retained for reproducibility.	Historical replay only where allowed.

- Downstream materializations, cached frames, methods and reports record the exact data/reference contract versions used.
- Impact analysis identifies affected artifacts after a mapping or scheme change.
- Current-state re-computation and historical replay are distinct operations. A later administrative reassignment does not silently rewrite a historical answer.

23. Typed Analytical Frames: Narrow Claim

Guarantee

A typed frame proves only declared compatibility of grain, dimensions, measures, time basis, versioned data/reference contracts and registered operators. It does not prove that the chosen method, peer group, attribution basis or interpretation is substantively appropriate.

```
frame: HospitalFunding
grain: [Hospital, FiscalYear]
measures:
  operating_funding:
    semantic_type: currency_amount
    aggregation: additive
reference_bindings:
  hospital_district:
    contract: health_facility_assignment@v12
    basis: administrative_location
time_basis: fiscal_year
```

24. Cross-Domain Conformed Interfaces and Semantic Bridges

- Use conformed interfaces such as AdministrativeArea x FiscalPeriod, Organization x FiscalPeriod and Person/Household identity only where semantics truly align.
- Every bridge declares attribution basis, lossiness, supported measures/methods, data/reference contract versions and blocked conditions.
- A bridge may be technically valid but method-inapplicable; Method Contracts must explicitly list compatible bridges/ attribution bases.
- Do not create a shared District x FY surface that hides service-catchment or scheme-allocation differences without declaring that loss.

25. Composition Governance and Structured STOP

STOP reason	Meaning	Default response
TYPE_INCOMPATIBLE	No legal typed composition.	Stop or choose registered transform.
MISSING_BRIDGE	Required cross-domain attribution/conformed interface absent.	Exploratory lane or bridge backlog.
MISSING_METHOD	No governed Method Contract matches the Question Specification.	Exploratory lane or method backlog.
METHOD_BINDING_AMBIGUOUS	Multiple materially different methods remain.	Clarify/attest; do not auto-select.
METHOD_OUT_OF_SCOPE	Selected contract fails applicability.	Redesign or stop.

STOP reason	Meaning	Default response
DATA_CONTRACT_UNHEALTHY	Required data/reference contract degraded beyond method tolerance.	Pause, downgrade or repair.
POLICY_BLOCK	Purpose/role/data combination is not authorized.	Stop; no workaround.
RESOURCE_LIMIT	Approved resource envelope insufficient.	Reduce scope or seek explicit higher budget.

Backlog rule

Only recurring MISSING_BRIDGE/MISSING_METHOD patterns with sufficient reuse, consequence or strategic value enter the governed promotion backlog. STOP telemetry is an input to prioritization, not an obligation to build a contract for every question.

26. Search Priority Is a Selection Mechanism, Not Evidence

SearchPriorityScore may order graph expansion but is not risk, evidence or probability. Under finite budgets it can nevertheless determine which candidates ever become visible, so it must be evaluated as a retrieval/ranking policy.

27. Search Policy Contract

```
search_policy:
  name: procurement_search_priority_v4
  objective: candidate_discovery
  exploitation_budget: 0.80
  exploration_budget: 0.20
  prohibited_operational_features:
    - protected_attribute_as_priority_signal
  diversity_controls:
    preserve_low_priority_sampling: true
    max_single_identifier_class_share: 0.40
  audit:
    require_discovery_coverage_certificate: true
    require_suppression_log: true
```

28. What Search Recall Can and Cannot Mean

Metric	When valid	Required label
Ground-truth candidate recall	Only when a defensible finite universe of qualifying candidates and labels exists.	REAL_WORLD_RECALL
Seeded/synthetic fixture detection rate	For known injected patterns and constructed red-team cases.	FIXTURE_DETECTION_RATE - not recall of unknown real candidates.
Reference-search recovery	Comparing live policy with a much larger-budget/stratified search over the same graph snapshot.	REFERENCE_RECOVERY - still not complete ground truth.
Exploration yield	Unique useful candidates found through reserved low-priority exploration.	EXPLORATION_YIELD
Adjudicated sample discovery rate	Within a sampled subset reviewed sufficiently to establish labels.	SAMPLED_ESTIMATE with confidence/coverage limits.

Rule

When the qualifying real-world candidate universe is unknown, the certificate must state true_candidate_recall: NOT_ESTIMABLE. A fixture detection rate such as 0.91 must never be displayed simply as "candidate recall 0.91".

29. Discovery Coverage Certificate

discovery_coverage_certificate:
 projection: procurement_network_v11
 search_policy: procurement_search_priority_v4
 snapshot: 2026-07-21
 node_budget: 10000000
 exploration_fraction: 0.20
 status: PARTIAL_SEARCH

 ground_truth:
 real_world_qualifying_universe_available: false
 true_candidate_recall: NOT_ESTIMABLE

 surrogate_evidence:
 seeded_fixture_detection_rate: 0.91
 larger_budget_reference_recovery_delta: 0.07
 exploration_unique_candidate_yield: 0.12
 adversarial_red_team_scenarios_passed: 34
 adversarial_red_team_scenarios_total: 40

 exposure_audit:
 slices_checked: [district, source_system, entity_type, identifier_class]
 material_disparity_flags: []
 suppressed_branch_sample_reviewed: true

 negative_statement:
 "No qualifying path was found within this graph snapshot, search policy and resource envelope. Real-world recall is not estimable."

30. Evaluation Without Complete Ground Truth

- Seeded fixtures test algorithmic capability against known patterns but are labeled as fixture tests, not real-world recall.
- Larger-budget or alternative-strategy reference searches estimate sensitivity to live budget/policy choices, not total graph truth.
- Random or stratified exploration samples provide adjudication outside the exploitation path and reduce selection bias.
- Adversarial red-team scenarios should include evasion patterns not used to tune the current search policy and should rotate over time.
- Suppression samples from common-hub rules are periodically reviewed to detect communities or regions systematically erased by pruning.
- Where statistically defensible labeled samples exist, report sample-based estimates with coverage and uncertainty; otherwise state NOT_ESTIMABLE.

31. Search Exposure and Fairness Governance

Test	Purpose
Expansion/exposure by operational slice	Detect geographic, source-system, entity-type or identifier-class blind spots.
Counterfactual perturbation	Test instability to non-substantive formatting/name/region proxies while preserving underlying graph evidence.
Exploration contribution	Measure the marginal candidate yield of reserved low-priority sampling.
Suppression audit	Inspect what hub/common-identifier rules removed.
Policy drift	Re-run validation after graph/source changes.
Review-capacity interaction	Check whether prioritization concentrates analyst workload on a narrow candidate class.

32. Relationship Evidence Models Remain Separate

- Search-policy features and evidence-model features may overlap, but model/version/semantics must remain separate.
- Evidence scoring begins only after a candidate is surfaced and may be domain-calibrated where labels permit.
- Search exposure bias is not repaired merely by calibrating downstream evidence models.
- No universal edge or path probability is permitted without a validated domain-specific probabilistic model.

33. Adaptive Search

Mode	Use	Governance requirement
Fast local	Interactive first-pass discovery.	Pre-approved small budget; exposure logging.
Deep adaptive	Authorized investigation of strong lead/complex chain.	Explicit higher resource budget and purpose.
Endpoint connection	Is A connected to B?	Declared endpoints and allowed edge families.
Global/offline	Recurring structural questions, index generation.	Batch governance, snapshot/version record.
Temporal flow	Ordered money/event flows.	Time/amount constraints and provenance.

34. Negative Search Statements

Required language

"No qualifying path found within this graph snapshot, policy and resource envelope." Never say "no relationship exists" unless the finite subgraph and search procedure genuinely establish that statement. When ground truth is unknown, also disclose that real-world candidate recall is not estimable.

35. Pre-Execution Authorization & Resource Gate

All expensive graph search or analytical execution is blocked until this gate passes.

Check	Examples
Purpose / actor / role	Is the requested analysis and graph projection authorized for this user and purpose?
Consequence class	Does this question require human pre-execution review regardless of method match?
Method binding	Is there a deterministic binding, confirmed default, or authorized attestation?
Data-access policy	Are requested datasets, cross-domain relationships and entity levels permitted?
Planned resource envelope	Maximum graph nodes/time, output rows, compute class, concurrency and cost ceiling.
Privacy/minimization	Can the task be answered with aggregated/tokenized data rather than person-level values?
Lane selection	Governed vs exploratory; exploratory lane cannot be used to bypass policy restrictions.

36. Plan Compilation and Post-Plan Computational Gate

After preflight authorization, the orchestrator compiles the intended governed plan without executing large scans/searches.

- Resolve active primitive, bridge, method and data/reference contract versions.
- Compile typed frame/operator plan and reject TYPE_INCOMPATIBLE or unhealthy-contract dependencies.
- Verify parameters are closed and resource estimates remain within the pre-approved envelope.
- Produce a signed/recorded Execution Plan ID.
- Only then release graph-search and analytical jobs.

37. Correct Execution Ordering

1. User/purpose authorization.
2. Canonical Question Specification and consequence classification.
3. Method routing, ambiguity resolution and/or binding attestation.
4. Pre-Execution Authorization & Resource Gate.
5. Resolve active data/reference contracts and compile typed/bridge plan.
6. Post-Plan Computational Gate validates composition and estimated execution envelope.
7. Execute graph search and/or SQL/OLAP/stats within the approved plan.
8. Run Methodological Gate diagnostics and robustness.
9. Apply evidence models only within allowed use.
10. Assemble Evidence Bundle and Discovery Coverage Certificate.
11. Evidentiary Gate constrains claim language.
12. Authorized human decision/action where required.

Ordering correction

A million-node search is never launched merely to discover later that the user's cross-domain operation is unauthorized, that the method binding is ambiguous, or that the resource envelope is disallowed. Preflight and plan compilation occur before expensive execution.

38. Gate Responsibilities

Gate	Runs when	Primary failures
Pre-Execution Authorization & Resource Gate	Before graph/analytics execution.	POLICY_BLOCK, METHOD_BINDING_AMBIGUOUS, HUMAN_APPROVAL_REQUIRED, RESOURCE_LIMIT.
Post-Plan Computational Gate	After governed plan compilation, before large execution.	TYPE_INCOMPATIBLE, DATA_CONTRACT_UNHEALTHY, invalid bridge/operator/version.
Methodological Gate	After data execution/diagnostics, before substantive interpretation.	REJECTED_METHOD, PASS_WITH_LIMITATIONS, OOD/diagnostic failures.

Gate	Runs when	Primary failures
Evidentiary Gate	After evidence bundle assembly.	INSUFFICIENT_EVIDENCE, LEAD_ONLY, CORROBORATED_ANOMALY, etc.

39. End-to-End Lifecycle v5

1. Receive natural-language question under an authenticated purpose and role.
2. Draft Canonical Question Specification; expose material inferences/ambiguities.
3. Confirm or clarify material fields where necessary.
4. Method Router enumerates eligible contracts and applicability conditions.
5. Resolve to deterministic binding, declared default, exploratory lane, or human method attestation.
6. Pre-Execution Gate validates authorization, consequence class, data access, privacy and resource envelope.
7. Compile governed plan from active primitives, frames, bridges and data/reference contracts.
8. Post-Plan Computational Gate validates typed composition and final resource estimates.
9. Execute graph search/analytics only within the recorded plan and budgets.
10. Run Methodological Gate diagnostics and robustness.
11. Apply domain evidence models according to calibration/allowed-use status.
12. Create Evidence Bundle plus honest Discovery Coverage Certificate.
13. Evidentiary Gate sets evidence level and permitted claim language.
14. Human review occurs before any consequential decision/action.

40. Audit and Reproducibility

- Persist original question and every Question Specification version.
- Record eligible Method Contract set, rejected candidates, binding status, defaults, clarifications and human attestation rationale.
- Record lane selection: governed or exploratory.
- Record data/reference contract versions, effective dates, lineage and health states.
- Record compiled Execution Plan ID, operator/bridge versions and pre/post gate decisions.
- Record graph snapshot, search policy, resource budget, exploration fraction, suppression log and Discovery Coverage Certificate.
- Record method diagnostics, robustness results, evidence-model versions and allowed-use status.
- Every material statement links to the corresponding Evidence Bundle and human-review state.

41. Operating Organization

Function	Primary responsibility	Can be combined in MVP?
Program/Product Lead	Scope, outcome metrics, stakeholder alignment, backlog and delivery governance.	Dedicated.
Platform/Architecture Lead	Reference architecture, interfaces, trust boundaries, technical decisions.	Dedicated.
Data/Analytics Engineering	Canonical data, semantic views, typed primitives, SQL/OLAP execution, lineage.	Team.
Metadata/Data-Quality Engineering	Data/reference contracts, effective dating, quality monitoring, catalog integration.	Can combine with data engineering initially.
LLM/Orchestration Engineering	Question specification, router integration, tool orchestration, prompt/evaluation harness.	1-2 engineers.
Graph/Search Engineering	Graph projection, search strategies, exposure logging and reference runs.	1-2 engineers.
Methodology/Statistics	Method families, applicability, diagnostics, robustness, evaluation.	At least 1 dedicated lead plus domain support.
Security/Privacy	Purpose enforcement, access controls, threat model, audit.	Dedicated or shared central security.
QA/SRE	Regression, reliability, performance, deployment and observability.	1-2 combined roles.
Domain Stewards	Meaning, policy context, reference data and method review.	Fractional by pilot domain.
Authorized Review Authority	High-consequence/novel method binding and decision governance.	Fractional but formally assigned.

42. Planning Staffing Envelope

The following is a planning envelope for a serious 6-9 month two-domain pilot. It is not a budget commitment and should be resized after discovery.

Role group	Indicative FTE	Notes
Program/Product + Architecture	2	One delivery lead and one platform architect.
Data/Analytics + Metadata/Quality	4-5	Canonical ingestion, semantic views, typed primitives, data/reference contracts, lineage.
LLM/Orchestration	1-2	Question specification, routing, orchestration, evaluation.
Graph/Search	1-2	Projection, search policy, exposure audit, performance.
Methodology/Statistics	1-2	Method families, diagnostics, applicability and experiment design.
Security/Privacy	1	May be shared with central team if service level is guaranteed.
QA/SRE	1-2	Automated regression, load, deployment, observability.
Pilot-domain stewards	0.3-0.5 each for 2 domains	Named accountable people, not an unbounded committee.
Authorized review authority	0.2-0.5	Depends on consequence profile and review volume.

Core capacity

A credible two-domain pilot is roughly 11-16 core FTE plus fractional domain/review capacity. Attempting the same scope with only a small LLM engineering team would shift unresolved work into hidden steward queues and invalidate the delivery assumptions.

43. Governance Throughput and Queue Controls

- Publish review queue age, median/95th-percentile lead time and backlog by governance tier.
- Set work-in-progress limits for Tier 2/3 contract reviews; do not start unlimited new domains simultaneously.
- Prioritize reviews using consequence, expected reuse, analyst demand and policy value - not first-come-first-served only.

- Provide an escalation path when a governance queue blocks an operationally urgent question.
- Track reviewer capacity as an explicit dependency in release planning.
- Pause domain expansion when the ratio of open governance items to available steward/reviewer capacity exceeds the agreed threshold.

44. Delivery Roadmap and Timeline

Period	Primary objective	Key deliverables	Exit criteria
Months 0-2	Foundation and operating model	Purpose model, two-lane policy, canonical Question Specification, basic method router, core semantic views, data-contract conventions.	One end-to-end low-consequence question routed and audited; staffing/owners named.
Months 2-5	Procurement governed pilot	Procurement primitives, graph projection, pre-exec gate, search policy/coverage certificate, 2-3 method contracts.	Repeatable governed workflow; no fake recall claims; review queue instrumented.
Months 5-8	Second domain + cross-domain bridge	Health or Welfare, effective-dated reference data, conformed interfaces, bridge contracts, exploratory-to-governed promotion.	At least one cross-domain governed workflow and one exploratory-only workflow correctly separated.
Months 8-11	Operational hardening	SRE, regression suites, lineage/impact, performance, security testing, steward SLAs.	Pilot availability/reliability targets met; backlog/lead time within agreed capacity.
Months 11-15	Selective expansion	Add 1-2 domains based on sponsor value and steward capacity, not all remaining domains at once.	Expansion approved only if governance queues and operational metrics remain healthy.
Beyond 15 months	Advanced/statistical methods	Forecasting/causal workflows only where explicit contracts and validation exist.	Method-specific acceptance criteria.

45. Recommended v0.1 MVP

- One primary pilot domain: Procurement/Finance; one secondary Health or Welfare workflow.
- Two operating lanes implemented and visibly labeled.
- Canonical Question Specification and deterministic Method Router with METHOD_BINDING_AMBIGUOUS behavior.
- 2-3 governed Method Contracts plus example rejected/ambiguous bindings.
- Core typed primitives and one or two conformed bridge templates.
- One effective-dated reference-data/data-contract case with automated validation and version-pinned replay.
- Pre-Execution Authorization & Resource Gate before graph/analytics execution.
- One operational graph with 10-20% configurable exploration budget, suppression log and Discovery Coverage Certificate showing true recall NOT_ESTIMABLE when appropriate.
- Isolated exploratory sandbox over approved semantic views; no direct promotion of exploratory outputs into consequential decisions.
- Operational dashboards for lane distribution, method-binding ambiguity, STOP reasons, governance queue age, data-contract health, search surrogate metrics and human review load.

46. Success Metrics

Area	Metrics
Question handling	Governed PASS %, PASS_WITH_LIMITATIONS %, exploratory-only %, ambiguity %, STOP %, human-attestation %.
Method binding	Wrong-binding incidents found in audit, ambiguous-binding rate, clarification rate, contract eligibility collisions.
Governance capacity	Tier lead time, queue age, review throughput, open items per steward FTE.
Composition	STOP reason distribution, bridge reuse, contract reuse, bespoke recipe rate.
Data/reference contracts	Health-state duration, breakage frequency, impacted artifacts, time to re-certification.
Search	Fixture detection rate, reference recovery delta, exploration yield, suppression audit findings, exposure drift; true recall only where defensible.
Evidence	Calibration/precision/false positives where labels exist, plus human overturn/disagreement rates.
Delivery	SLOs, cost per governed workflow, analyst cycle time, promotion rate from exploratory to governed.

47. Cost and Capacity Controls

- Every graph search has a pre-approved resource envelope before execution; higher budgets require explicit escalation.
- Exploratory workloads receive stricter row/time/concurrency caps and may be queued separately from governed workloads.
- Method/bridge authoring capacity is planned per release; domain expansion cannot assume infinite steward throughput.
- Contract count, test burden and maintenance cost are tracked alongside question coverage to prevent hidden sprawl.
- Sunset unused contracts and archive low-value graph projections.

48. Worked Example: Ambiguous Hospital Question

Question: "Are hospitals receiving more money without getting better?"

1. The LLM drafts a Question Specification but flags two ambiguities: "more money" could mean operating funding or total funding, and "getting better" could mean output volume, quality outcomes or efficiency.
2. The router finds three eligible Method Contracts corresponding to funding-output, funding-quality and efficiency-change questions.
3. Because the eligible methods imply materially different outcomes, the status is METHOD_BINDING_AMBIGUOUS.
4. The system asks the analyst to choose/clarify the intended outcome family; it does not silently select the funding-output method.
5. After confirmation, the selected Method Contract is bound and the governed path proceeds.
6. If the analyst declines to clarify, the question may be explored in the sandbox, but the result remains EXPLORATORY_NOT_CERTIFIED.

Why this matters

A wrong-but-internally-consistent Method Contract can no longer "sail through every gate" merely because the LLM chose it. The architecture still cannot prove the analyst's intent, but it makes ambiguity and binding evidence first-class objects.

49. Worked Example: Cross-Domain Hospital and Welfare

1. Question: "Do districts with increasing hospital funding also show deteriorating welfare coverage?"
2. The Question Specification requires an attribution basis for hospital-to-district mapping and a welfare scheme/scheme-group definition.
3. If a governed CrossDomainTrendAssociation Method Contract supports administrative-location attribution and the required scheme group, the router binds it.
4. If the district attribution is unresolved or multiple scheme groupings materially change the meaning, the binding is ambiguous and requires clarification.
5. If no governed method exists, the analyst may use the exploratory lane to investigate read-only conformed views. The output is labeled exploratory.
6. If the pattern becomes recurrent/high-value, the exploratory workflow is promoted through Tier 2 review into a reusable Method Contract/bridge.

50. Worked Example: Procurement Search Without False Recall

A procurement graph search runs under a one-million-node budget after pre-execution authorization.

- The live policy detects 91% of seeded/red-team fixtures. The certificate labels this seeded_fixture_detection_rate: 0.91.
- A larger-budget reference search recovers 7% additional candidates relative to the live policy.
- Reserved exploration contributes 12% unique surfaced candidates.
- No defensible universe of all real collusive/undisclosed-ownership candidates exists, so true_candidate_recall is NOT_ESTIMABLE.
- The analyst-facing summary may state that the search is partial and provide the surrogate coverage evidence; it may not say "the search has 91% recall."

51. Residual Risks and Honest Limitations

Residual risk	Why it remains	Mitigation / posture
Method binding can still be wrong	Canonicalization and human clarification do not reveal the user's latent intent perfectly.	Expose ambiguity, record binding evidence, audit wrong-binding incidents, require attestation for high consequence.

Residual risk	Why it remains	Mitigation / posture
Governance bottleneck remains	Governed reuse requires stewardship and review.	Two-lane model, method families, capacity metrics, WIP limits, staged domain expansion.
Exploratory sandbox may produce misleading analysis	Broader freedom increases methodological risk.	Strong EXPLORATORY_NOT_CERTIFIED labeling, isolation, row/resource limits, no direct consequential use.
True graph-search recall is usually unknown	Undisclosed/adversarial targets lack complete ground truth.	Report NOT_ESTIMABLE; use surrogate diagnostics and red-team/reference searches.
Standard data contracts still require operational ownership	Tools do not maintain semantics automatically.	Named owners, change notices, validation jobs, impact analysis and SLAs.
Conformed dimensions can oversimplify	Cross-domain alignment can erase meaningful attribution differences.	Bridge lossiness + method compatibility + explicit attribution basis.
Staffing assumptions may be underfunded	Government hiring/procurement cycles can exceed technical timelines.	Treat capacity as a release dependency; reduce domains/scope rather than hiding queue time.
LLM over-interpretation	Fluency can exceed evidence and method assurance.	Evidence-level claim templates, explicit lane/limitations, human review.

52. Non-Negotiable Architecture Rules

- No claim that typed composition proves methodological appropriateness.
- No claim that Method Contract conformance proves the LLM selected the right method. Method binding is a separate assurance problem.
- No silent auto-selection when multiple materially different Method Contracts are eligible.
- No consequential use of exploratory-lane outputs without governed promotion or explicit authorized exceptional process.
- No expensive graph/analytical execution before the Pre-Execution Authorization & Resource Gate passes.
- No real-world "candidate recall" metric without a defensible labeled candidate universe. Seeded fixture detection must be labeled as such.
- No invented novelty claims for SCD/effective dating, data contracts, data quality or lineage; use standard terminology and existing tooling patterns where practical.
- No timeless administrative mapping when the relationship can change; preserve effective-dated history and attribution basis.
- No governed composition against data/reference contracts outside their permitted health state.
- No universal graph relationship score/path probability across domains without validation.
- No graph search-priority score presented as risk, evidence or probability.
- No domain expansion without named steward/reviewer capacity and measured queue health.
- Every material result records its lane, question specification, method binding, execution plan, data/reference versions, search coverage statement, evidence level and human-review state.

Conclusion

Architecture v5 changes the claim from "a government-wide governed system can make most analytical judgment mechanically safe" to a narrower and more operationally credible proposition: a government-wide substrate can support broad exploration while reserving high assurance for analyses whose intent, method, data contracts, computation and evidence have each passed the appropriate trust boundary.

The remaining hard problems - interpreting intent, acquiring domain knowledge, discovering unknown graph patterns and staffing governance - are not treated as implementation details. They are explicit design constraints with separate operating modes, statuses, capacity assumptions and failure behavior.

Appendix A. Question Specification Minimum Schema

```
question_spec:
  original_question: ...
  purpose: ...
  consequence_class: low | medium | high
  population: {entity_type, inclusion_rule}
  exposure: {semantic_role, metric_family}
  outcomes: [...]
  comparison_request: ...
  time: {basis, period}
  attribution_basis: ...
  causal_language_requested: true | false
  user_confirmed_fields: [...]
  defaulted_fields: [...]
  ambiguity_flags: [...]
  version: ...
```

Appendix B. Method Binding Record

```
method_binding:
  question_spec_version: ...
  eligible_contracts: [...]
  rejected_contracts:
    - contract: ...
      reason: ...
  selected_contract: ...
  status: BINDING_DETERMINISTIC | BINDING_DEFAULTED |
          METHOD_BINDING_AMBIGUOUS | MISSING_METHOD |
          METHOD_OUT_OF_SCOPE | HUMAN_METHOD_ATTESTATION
  default_rules_applied: [...]
  clarification_record: ...
  human_attestation:
    approver: ...
    rationale: ...
    limitations: ...
```

Appendix C. Method Contract Minimum Schema

```
method_contract:
  id: ...
  question_classes: [...]
  permitted_consequence_classes: [...]
  semantic_roles:
    required: [...]
  applicability:
    populations: [...]
    time_constraints: ...
    attribution_bases: [...]
    data_contract_requirements: [...]
  peer_templates: [...]
  estimands: [...]
  denominators: [...]
  diagnostics: [...]
  robustness: [...]
  interpretation_limits: [...]
  bridge_compatibility: [...]
  OOD_triggers: [...]
  regression_cases: [...]
  counterexamples: [...]
  version: ...
```

Appendix D. Data/Reference Contract Minimum Schema

```
data_contract:
  id: ...
  dataset_or_reference: ...
  business_validity:
    effective_from: ...
    effective_to: ...
  source_version: ...
  key_assertions: [...]
  cardinality_assertions: [...]
  referential_integrity: [...]
  freshness_threshold: ...
  completeness_threshold: ...
  semantic_version: ...
  health: ACTIVE | DEGRADED | QUARANTINED | RETIRED
  owner: ...
  lineage: [...]
  impacted_artifacts: [...]
```

Appendix E. Structured STOP Taxonomy

Code	Meaning
METHOD_BINDING_AMBIGUOUS	Several materially different governed methods remain plausible.
MISSING_METHOD	No governed Method Contract supports the specification.
METHOD_OUT_OF_SCOPE	Contract exists but applicability conditions fail.
TYPE_INCOMPATIBLE	Typed composition/operator mismatch.
MISSING_BRIDGE	No governed cross-domain mapping/interface exists.
DATA_CONTRACT_UNHEALTHY	Required data/reference contract cannot support governed execution.
POLICY_BLOCK	Purpose/role/data combination not authorized.
RESOURCE_LIMIT	Execution exceeds pre-approved resource envelope.
HUMAN_APPROVAL_REQUIRED	Governed path exists but authorized pre-execution judgment is mandatory.
EXPLORATORY_ONLY	Question may be researched in sandbox but lacks governed assurance.

Appendix F. Discovery Coverage Certificate Minimum Schema

```
discovery_coverage:
  graph_snapshot: ...
  search_policy: ...
  resource_envelope: ...
  status: COMPLETE_FINITE_SUBGRAPH | PARTIAL_SEARCH
  ground_truth:
    available: true | false
    true_candidate_recall: value | NOT_ESTIMABLE
  surrogate_evidence:
    seeded_fixture_detection_rate: ...
    reference_search_recovery_delta: ...
    exploration_yield: ...
    adversarial_red_team: ...
  exposure_audit: ...
  suppression_audit: ...
  negative_statement: ...
```

Appendix G. Resourcing Assumptions

- 11-16 core FTE is the planning envelope for a serious two-domain pilot; fractional domain/review roles are additional.
- Review lead-time targets assume named reviewers with protected capacity; they are not achievable if stewardship is an unpaid side duty.
- Domain expansion is gated by queue health and platform reliability, not by the calendar alone.
- Procurement/hiring/security accreditation delays should be tracked as program risks separately from engineering progress.

- Cost estimates should be produced after discovery using actual source-system count, graph size, concurrency, data sensitivity and infrastructure policy.

Appendix H. Migration from v4

- Replace "Method Contract selection by LLM" with Canonical Question Specification + deterministic Method Router + binding statuses.
- Add Interpretive/Method-binding validity as a separate assurance dimension.
- Split the single governed path into Governed Decision-Support and Exploratory Research lanes.
- Rename Temporal Dependency Registry implementation language to effective-dated reference/master data + data contracts + lineage/impact, while preserving effective-date and historical replay requirements.
- Replace `candidate_recall_on_validation_suite` with explicitly labeled fixture detection/reference recovery metrics; set true recall to `NOT_ESTIMABLE` when labels are unavailable.
- Move authorization/resource checks before any expensive graph or analytical execution; retain a post-plan Computational Gate.
- Add staffing envelope, review-capacity metrics, governance WIP controls and staged timeline.
- Revise mission language from universal governed answerability to government-wide substrate with tiered assurance.

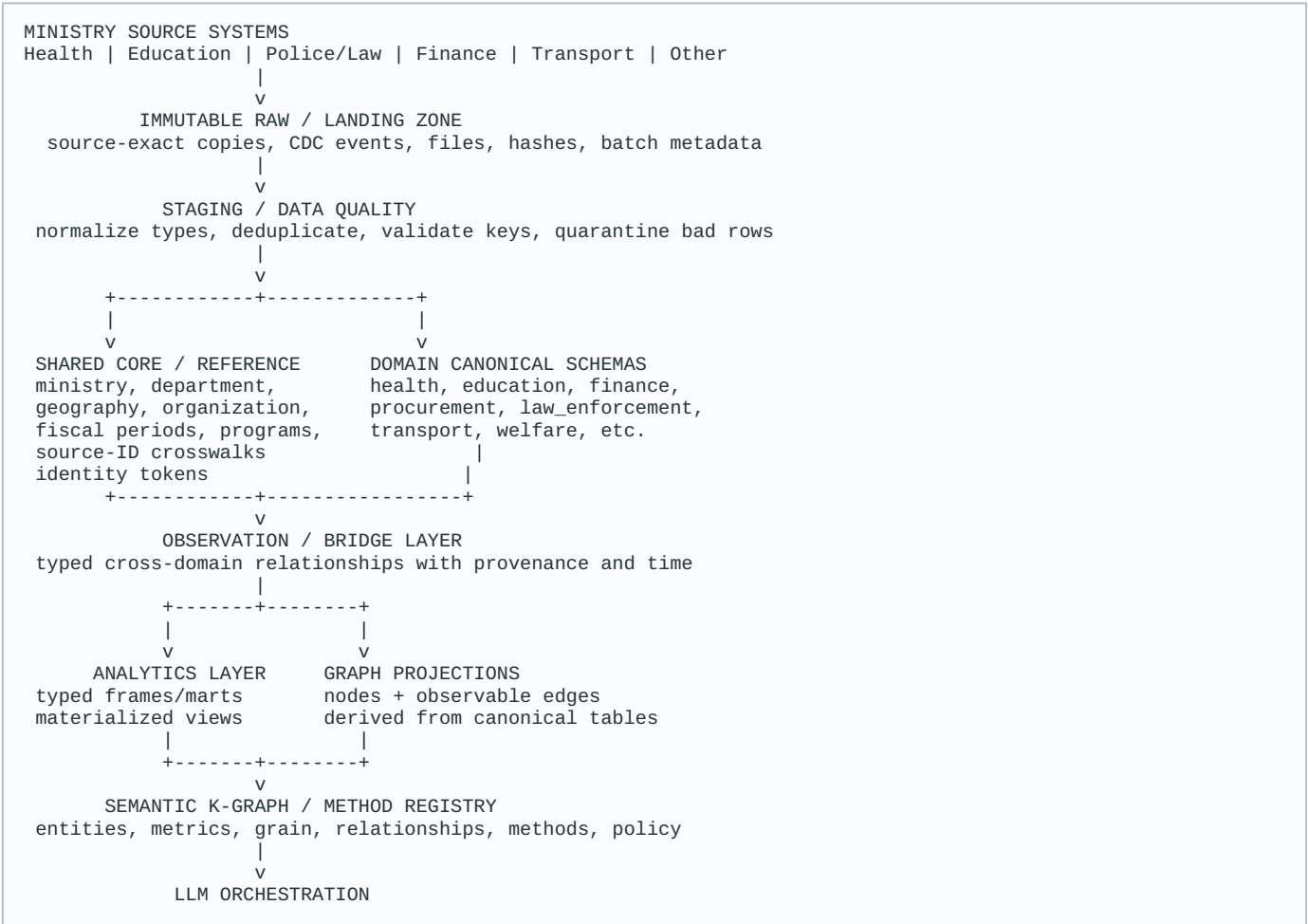
Appendix I. Central Government Data Platform and Canonical Database Design

This appendix defines how data migrated from Health, Education, Law Enforcement, Finance, Procurement, Transport and future ministries should be organized so the analytical platform, governed SQL layer and K-Graph can operate consistently. The central design principle is **not** to copy every ministry database into one giant relational schema. The platform should preserve source fidelity, create a common identity/reference spine, keep domain facts typed at their natural grain, and publish graph/analytical projections from canonical data.

Recommended platform shape

The "central database" should be treated as a central **data platform**: immutable raw storage + canonical columnar warehouse/lakehouse + tightly controlled identity/reference store + derived graph projections + semantic metadata. A single general-purpose RDBMS should not be forced to perform every role.

I.1 End-to-End Data Layout



I.2 Logical Schema / Namespace Layout

Schema / zone	Purpose	LLM governed-lane access
raw_health, raw_education, raw_law, raw_finance, raw_transport	Immutable source-exact snapshots, CDC records and file payload metadata. Preserve source identifiers and source timestamps exactly.	No direct governed analytical access.
stg_*	Type normalization, source-specific cleaning, deduplication, code mapping and data-quality quarantine.	No direct governed analytical access.
core	Cross-government conformed master/reference dimensions: ministries, departments, geography, organizations, programs, fiscal/calendar periods and non-sensitive entity registry.	Through governed semantic objects and approved joins.

Schema / zone	Purpose	LLM governed-lane access
secure_identity	Restricted PII/identity crosswalks and tokenization. Separates names/identifiers from general analytical tables.	Only through authorized identity service or purpose-specific views.
health, education, finance, procurement, law_enforcement, transport	Canonical domain dimensions, transactions, events and snapshots at declared natural grain.	Normally through registered primitives/ views; limited exploratory access through approved semantic views.
bridge	Typed cross-domain observations: entity-location, organization-role, account-use, ownership, program participation and approved entity-resolution results.	Only relationships permitted by policy/ purpose.
analytics	Governed star-schema marts, materialized summaries and typed analytical frames optimized for SQL/OLAP.	Primary governed analytical surface.
graph	Derived node/edge projection tables used to load the operational graph engine. Not the source of truth.	Through registered graph projections/ search policies.
meta	Source catalog, semantic/K-Graph registry, data contracts, metric/frame definitions, lineage, method/bridge contracts and versions.	Used by the LLM/routers to discover legal analytical objects.
audit	Ingestion logs, access logs, query/execution plans, model/ method versions, policy decisions and result lineage.	Read for audit/reproducibility, not ordinary analysis.

I.3 Key Strategy and Mandatory Record Metadata

Do not use a ministry's source primary key as a government-wide key. Every canonical entity should receive a stable platform identity while source keys remain preserved in crosswalk tables.

Field pattern	Rule	Reason
*_sk BIGINT	Warehouse surrogate key used for high-volume joins and fact tables.	Compact and fast for analytics.
*_uid UUID	Stable globally unique platform identifier for cross-system references, APIs and graph nodes.	Portable identity independent of warehouse row numbering.
source_system_id + source_record_id	Preserved in a metadata/xref table rather than treated as the canonical identity.	Source IDs can collide or be reassigned across ministries.
effective_from / effective_to	Required for time-varying reference relationships and dimensional history.	Prevents current assignments from rewriting historical analysis.
event_time / transaction_time	Stored on facts where a real-world event/transaction occurred.	Supports correct temporal analysis.
source_updated_at	Timestamp from source system where available.	Supports CDC and conflict resolution.
loaded_at / ingestion_batch_id	Platform ingestion metadata on every canonical load.	Reproducibility and rollback.
record_hash	Hash of normalized source payload or business fields.	Change detection and reconciliation.
data_classification	Classification should be attached through metadata/policy, and row/column tags where needed.	Purpose enforcement, masking and minimization.

Do not build a universal EAV database

A single table such as *entity_id*, *attribute_name*, *value* appears flexible but destroys strong typing, grain semantics, constraints, performance and meaningful SQL generation. Use a small common identity/reference spine plus typed domain tables.

I.4 Shared Core Tables

Table	Natural grain	Important columns	Use
core.ministry	One ministry version	ministry_sk, ministry_uid, code, name, effective_from/to	Top-level ownership and policy boundary.
core.department	One department version	department_sk, ministry_sk, code, name, effective_from/to	Department hierarchy.

Table	Natural grain	Important columns	Use
core.organization	One organization version	organization_sk, organization_uid, organization_type, legal_name, department_sk, status, effective_from/to	Government offices, vendors, institutions and other organizations. Domain-specific attributes remain in domain tables.
core.administrative_area	One geographic unit version	area_sk, area_uid, area_type, parent_area_sk, code, name, effective_from/to	State/district/local-body hierarchy with historical boundary versions.
core.program	One scheme/program version	program_sk, program_uid, owner_department_sk, code, name, program_type, effective_from/to	Government schemes, grants and programs.
core.fiscal_period	One governed fiscal period	fiscal_period_sk, fiscal_year, period_type, start_date, end_date, calendar_version	Common fiscal alignment.
core.calendar_date	One date	date_sk, calendar_date, month, quarter, fiscal_period_sk, holiday flags	Consistent date/fiscal joins.
core.entity_registry	One global entity identity	entity_uid, entity_type, canonical_domain, status	Identity spine only; not a generic attribute store.
meta.source_system	One source application/version	source_system_id, ministry_sk, owner, system_name, environment, classification	Source provenance.
meta.source_record_xref	One source record to canonical entity mapping	source_system_id, source_record_id, entity_uid, valid_from/to, match_method, match_version	Source-to-platform identity crosswalk.

I.5 Identity and Sensitive Person Data

Person-level identity is the highest-risk cross-ministry join surface. The general warehouse should use tokenized person keys; direct identifiers should be isolated in **secure_identity** and released only under an authorized purpose.

Table	Purpose
secure_identity.person_identity	Encrypted canonical identity record; restricted legal name/DOB and approved identity attributes. Not available to ordinary analytical SQL.
secure_identity.identifier	Encrypted/tokenized identifiers with identifier_type, issuer, validity and verification status.
core.person_token	Opaque person_uid/person_sk visible to approved analytical tables without exposing direct identifiers.
secure_identity.identity_match_audit	Evidence, model/rule version, reviewer state and confidence for record-linkage decisions.

- Never join ministries merely because two source systems contain the same name or phone number.
- Cross-domain person matching must pass the identity-resolution and policy layer.
- Law-enforcement roles such as victim, witness, complainant, person of interest, suspect and accused remain explicit role records; they are never collapsed into one generic adverse status.

I.6 Health Canonical Tables

Table	Declared grain	Examples of measures/attributes
health.facility_dim	One facility version	facility_sk, organization_sk, facility_type, referral_level, ownership_class, effective_from/to
health.facility_area_history	Facility x attribution basis x effective period	facility_sk, area_sk, attribution_basis, effective_from/to, source
health.encounter_fact	One healthcare encounter	encounter_uid, facility_sk, person_sk/token where permitted, encounter_type, admission/discharge timestamps, case-mix group
health.service_activity_fact	Facility x date/period x service type	admissions, outpatient_visits, surgeries, deliveries, occupied_bed_days
health.funding_fact	Facility/program x fiscal period x funding transaction/category	operating_funding, capital_funding, grant_amount, source_department
health.capacity_snapshot_fact	Facility x snapshot date	licensed_beds, staffed_beds, ICU_beds, operating_theatres
health.staffing_snapshot_fact	Facility x staff category x snapshot period	sanctioned_posts, filled_posts, FTE
health.outcome_fact	Facility/clinical cohort x period x metric	mortality, readmission, wait-time components, quality numerator/denominator

Health design note

Keep event-level clinical/person data in a restricted partition or enclave when possible. Publish aggregated facility-period facts for most government performance analytics. Ratios such as occupancy or mortality should retain governed numerator/denominator components rather than only the final percentage.

I.7 Education Canonical Tables

Table	Declared grain	Examples
education.institution_dim	One school/college/institution version	institution_sk, organization_sk, institution_type, management_type, area_sk, effective dates
education.enrollment_fact	Institution x academic period x grade/program x student category	enrolled_count, new_enrollment_count, transfer_count
education.attendance_fact	Institution x date/period x grade/program	present_student_days, possible_student_days, absence_count
education.assessment_fact	Institution/student-token where permitted x assessment x subject	score, grade, pass indicator, assessment version
education.staffing_snapshot_fact	Institution x staff category x snapshot date	sanctioned_posts, filled_posts, teacher_FTE
education.funding_fact	Institution/program x fiscal period x funding category	allocation, release, expenditure
education.infrastructure_snapshot_fact	Institution x snapshot date x infrastructure item	classrooms, labs, toilets, devices, condition status

I.8 Finance and Procurement Canonical Tables

Table	Declared grain	Examples
finance.budget_allocation_fact	Department/program/account head x fiscal period x allocation event	original_budget, supplementary_allocation, revised_budget
finance.expenditure_fact	One posted expenditure/accounting line	department_sk, program_sk, account_head, amount, posting_date, voucher reference
finance.payment_fact	One payment/disbursement event	payer_org_sk, payee_entity_uid, amount, currency, payment_date, instrument/account token
finance.revenue_receipt_fact	One receipt/collection posting	revenue_head, payer token/entity, amount, receipt_date
procurement.contract_dim	One contract version	contract_uid, department_sk, category, procurement_method, start/end, awarded_vendor_sk, status
procurement.bid_fact	Contract/tender x bidder x bid submission/version	bid_amount, submission_time, qualification status
procurement.award_fact	One award decision	contract_uid, vendor_sk, awarded_amount, award_date, decision status
procurement.amendment_fact	One contract amendment	amendment_type, amount_change, duration_change, approval_date
procurement.performance_fact	Contract x reporting period	milestone status, delay_days, delivered_amount/quantity, completion status

Vendor identity

Represent a vendor primarily as a role of **core.organization**. Do not create separate incompatible vendor identities inside every department. Source vendor numbers belong in the source-record crosswalk.

I.9 Law-Enforcement Canonical Tables

Table	Declared grain	Examples
law_enforcement.agency_unit_dim	One agency/station/unit version	organization_sk, unit_type, jurisdiction_area_sk, effective dates

Table	Declared grain	Examples
law_enforcement.incident_fact	One reported incident/event	incident_uid, incident_type, occurred_at, reported_at, location_area_sk, source_unit_sk
law_enforcement.case_dim	One case/investigation version	case_uid, case_type, opening/closing dates, responsible_unit_sk, status
law_enforcement.case_party_role	Case x person/entity x role x effective period	party_entity_uid, role_type, role_status, effective_from/to, source/provenance
law_enforcement.action_event_fact	One authorized case action/event	action_type, action_time, responsible_unit, legal_authority reference
law_enforcement.evidence_metadata	One evidence-item metadata record	evidence_uid, evidence_type, custody/status metadata, restricted object-store reference
law_enforcement.custody_event_fact	One custody-status event where legally permitted	person_token, event_type, start/end time, authority reference

Law-enforcement boundary

The central analytical warehouse should not become an unrestricted copy of operational evidence systems. Highly sensitive documents, media and investigative content should remain in controlled evidence/object repositories; the warehouse stores only the metadata and analytical fields authorized for the platform. Graph projections must preserve precise legal/operational roles and purpose restrictions.

I.10 Transport Canonical Tables

Table	Declared grain	Examples
transport.route_dim	One route/version	route_uid, operator_org_sk, mode, origin/destination, effective dates
transport.stop_station_dim	One stop/station version	stop_uid, mode, area_sk, geospatial reference, effective dates
transport.service_trip_fact	One scheduled/actual service trip	route_sk, vehicle_sk, scheduled/actual times, delay, cancellation indicator
transport.ridership_fact	Route/stop/service x period	boardings, alightings, passenger_km, fare revenue
transport.vehicle_asset_dim	One vehicle/asset version	asset_uid, operator_org_sk, asset_type, commissioning date, status
transport.vehicle_status_snapshot_fact	Vehicle x snapshot date	availability, odometer, maintenance status
transport.road_segment_dim	One road segment version	segment_uid, road_class, owner_org_sk, area references
transport.traffic_count_fact	Road segment x observation period x vehicle class	vehicle_count, average_speed, congestion measures
transport.incident_fact	One transport/road incident	segment/route, occurred_at, incident_type, severity category
transport.maintenance_work_order_fact	One asset/segment work order	work_type, planned/actual dates, cost, contractor/vendor

I.11 Reusable Pattern for Additional Ministries

New ministries should not invent a completely new modeling philosophy. Each domain should map its data into four standard table patterns:

Pattern	What one row means	Examples
Entity/dimension history	One business entity version valid during an effective period.	Hospital, school, route, contract, government unit.
Transaction/event fact	One real-world transaction or event.	Payment, encounter, bid, incident, trip.
Periodic snapshot fact	One entity/category at a regular reporting point.	Staffing by month, bed capacity, school infrastructure, fleet availability.

Pattern	What one row means	Examples
Aggregate activity fact	One declared entity/time/category grain with additive components.	Hospital services by month, enrollment by academic period, route ridership by day.

Grain first

Every canonical fact table must have a written one-row grain statement before migration. A table named *hospital_data* containing funding, patient rows, staff and district summaries at mixed levels must be split before it enters the governed analytical layer.

I.12 Cross-Domain Bridge / Observation Tables

Cross-ministry analysis and graph search require common observable relationships. These should be represented as typed observations with source, time and role - not as vague inferred links.

Table	One row represents	Important fields
bridge.entity_location_observation	An entity associated with a location/address under a declared role.	entity_uid, location_uid/area_sk, location_role, observed_at, valid_from/to, source_record
bridge.person_organization_role	A person has a declared role in an organization.	person_uid, organization_uid, role_type, start/end, source, verification status
bridge.organization_relationship	A typed legal/administrative relationship between organizations.	from_org_uid, to_org_uid, relationship_type, effective dates, source
bridge.account_usage_observation	An entity is observed using/receiving through a tokenized financial account.	entity_uid, account_token, usage_role, observed_at, valid dates, source
bridge.asset_ownership_observation	An entity owns/controls/uses an asset under a declared role.	entity_uid, asset_uid, ownership_role, effective dates, source
bridge.program_participation	An entity/organization participates in a government program.	entity_uid, program_sk, participation_role, start/end, source
bridge.entity_match	An approved identity-resolution assertion between source identities.	left_entity, right_entity, match_status, method/model version, reviewer state

Graph rule

Store observations such as "Organization A uses Account X" and "Person P is director of Organization A". Do not store a source-of-truth edge called "Organization A suspiciously_related_to Organization B". Interpretive relationships belong in leads/findings with evidence provenance.

I.13 K-Graph Integration: Semantic Graph vs Operational Entity Graph

Graph	What it contains	Where it comes from
Semantic K-Graph	Concepts, table/view meanings, grains, metrics, dimensions, allowed transformations, data contracts, methods, policies and lineage.	meta registry + steward-authored semantic objects.
Operational Entity Graph	Selected real-world entity nodes and observable relationship edges used for traversal/network investigation.	core/domain/bridge canonical tables projected through registered graph projections.

The semantic graph tells the LLM *how government data is structured and what operations are legal*. The operational entity graph tells the system *which real-world entities are observed to be connected*.

I.13.1 Semantic Registry Tables

Table	Purpose
meta.semantic_entity	Hospital, School, Vendor, Contract, PersonToken, Route, District, Program, etc.
meta.semantic_property	Typed properties and their physical/canonical mappings.
meta.semantic_relationship	Declared relationships such as Hospital located_in District or Contract awarded_to Organization.
meta.metric	Metric definition, numerator/denominator, aggregation behavior, units, versions.
meta.frame_type	Typed analytical frame grain, measures, time basis and compatible operations.

Table	Purpose
meta.primitive	Registered SQL/OLAP capability producing a typed frame.
meta.bridge_contract	Governed cross-domain attribution/mapping contract.
meta.method_contract	Question classes, applicability, peer/baseline rules, diagnostics and interpretation limits.
meta.data_contract	Schema, key, quality, freshness, effective-dating and health requirements.

I.13.2 Derived Graph Projection Tables

Table	Purpose
graph.node_projection	Stable graph node identifiers mapped to entity_uid, entity_type, display classification and graph projection version.
graph.edge_projection	Typed observable edges with from/to nodes, edge_type, valid/observed time, source-table/source-row provenance and access policy.
graph.projection_snapshot	Projection version, source watermarks, build time, data coverage and policy version.

I.14 Analytics / Typed Frame Layer

Governed LLM execution should target curated analytical views/materialized tables rather than repeatedly joining raw ministry tables. These objects have explicit grain, metric behavior and versions.

Example analytical object	Declared grain	Typical use
analytics.health_hospital_funding_year	Hospital x FiscalYear x FundingCategory	Funding trend and rollup analysis.
analytics.health_hospital_output_year	Hospital x FiscalYear x ServiceType	Admissions/surgeries/outpatient output.
analytics.education_institution_performance_year	Institution x AcademicYear x Metric	Enrollment, attendance and assessment comparisons.
analytics.finance_department_spend_month	Department x Program x Month x AccountHead	Budget/expenditure monitoring.
analytics.procurement_vendor_awards_year	Vendor x Department x Category x FiscalYear	Award concentration/peer analysis.
analytics.transport_route_performance_month	Route x Month	Ridership, reliability, cost/revenue analysis.

Governed SQL boundary

In the governed lane, the LLM should usually select registered analytical primitives/views whose SQL is reviewed and versioned. The exploratory lane may receive broader read-only semantic views, but never raw unrestricted production credentials.

I.15 Example Physical DDL Patterns

The following is illustrative SQL. Exact data types and partition syntax should be adapted to the selected warehouse/lakehouse engine.

I.15.1 Source System and Crosswalk

```
CREATE TABLE meta.source_system (
  source_system_id    BIGINT PRIMARY KEY,
  ministry_sk         BIGINT NOT NULL,
  system_code         VARCHAR(100) NOT NULL,
  system_name         VARCHAR(300) NOT NULL,
  source_owner        VARCHAR(300),
  data_classification VARCHAR(50) NOT NULL,
  active_from         DATE NOT NULL,
  active_to           DATE,
  UNIQUE (ministry_sk, system_code)
);

CREATE TABLE meta.source_record_xref (
  source_system_id    BIGINT NOT NULL,
  source_record_id    VARCHAR(300) NOT NULL,
  entity_uid          UUID NOT NULL,
  entity_type         VARCHAR(80) NOT NULL,
  valid_from          TIMESTAMP NOT NULL,
  valid_to            TIMESTAMP,
  match_method        VARCHAR(80) NOT NULL,
  match_version       VARCHAR(80),
  loaded_at           TIMESTAMP NOT NULL,
  PRIMARY KEY (source_system_id, source_record_id, valid_from)
);
```

I.15.2 Organization and Historical Attributes

```
CREATE TABLE core.organization (
  organization_sk      BIGINT PRIMARY KEY,
  organization_uid     UUID NOT NULL,
  organization_type    VARCHAR(80) NOT NULL,
  legal_name          VARCHAR(500),
  owner_department_sk BIGINT,
  status_code         VARCHAR(50),
  effective_from       DATE NOT NULL,
  effective_to         DATE,
  loaded_at           TIMESTAMP NOT NULL,
  ingestion_batch_id   BIGINT NOT NULL,
  record_hash          VARCHAR(128) NOT NULL,
  UNIQUE (organization_uid, effective_from)
);
```

I.15.3 Health Activity Fact

```
CREATE TABLE health.service_activity_fact (
  activity_sk          BIGINT PRIMARY KEY,
  facility_sk          BIGINT NOT NULL,
  period_sk           BIGINT NOT NULL,
  service_type_code    VARCHAR(80) NOT NULL,
  admissions_count     BIGINT,
  outpatient_visits    BIGINT,
  surgeries_count      BIGINT,
  occupied_bed_days    NUMERIC(18,2),
  source_system_id     BIGINT NOT NULL,
  source_updated_at    TIMESTAMP,
  loaded_at           TIMESTAMP NOT NULL,
  ingestion_batch_id   BIGINT NOT NULL,
  record_hash          VARCHAR(128) NOT NULL,
  UNIQUE (facility_sk, period_sk, service_type_code, source_system_id)
);
```

I.15.4 Observable Relationship for Graph Projection

```
CREATE TABLE bridge.account_usage_observation (
  observation_uid      UUID PRIMARY KEY,
  entity_uid           UUID NOT NULL,
  account_token        VARCHAR(200) NOT NULL,
  usage_role           VARCHAR(80) NOT NULL,
  observed_at          TIMESTAMP,
  valid_from           TIMESTAMP,
  valid_to             TIMESTAMP,
  source_system_id     BIGINT NOT NULL,
  source_record_id     VARCHAR(300) NOT NULL,
  verification_status  VARCHAR(50),
  loaded_at           TIMESTAMP NOT NULL
);
```

I.15.5 Derived Graph Edge

```
CREATE TABLE graph.edge_projection (
  projection_version    VARCHAR(80) NOT NULL,
  edge_uid             UUID NOT NULL,
  from_node_uid        UUID NOT NULL,
  edge_type            VARCHAR(100) NOT NULL,
  to_node_uid          UUID NOT NULL,
  observed_at          TIMESTAMP,
  valid_from           TIMESTAMP,
  valid_to             TIMESTAMP,
  source_table         VARCHAR(200) NOT NULL,
  source_row_uid       VARCHAR(300) NOT NULL,
  access_policy_id     VARCHAR(100) NOT NULL,
  PRIMARY KEY (projection_version, edge_uid)
);
```

I.16 Migration Process from Existing Government Servers

1. **Inventory and classify.** Register every source system/table/file, owner, refresh pattern, data classification, retention rule and authoritative status.
2. **Define the target grain before copying into canonical tables.** For each source table state exactly what one row represents and whether it is an entity version, transaction, event, snapshot or aggregate.
3. **Land source-exact data first.** Copy snapshots/CDC into immutable raw storage with source path/table, extraction time, row count, checksums and source watermark.
4. **Profile and quarantine.** Measure nulls, duplicates, invalid dates, key collisions, orphan references and code-set drift. Bad records go to a quarantine table; do not silently coerce them.
5. **Normalize source-specific codes.** Map ministry-local district, facility, school, vendor, program and fiscal codes to governed reference tables while retaining the original values.
6. **Resolve identities.** Assign stable organization/facility/institution/entity identities. Person resolution occurs only through the protected identity service.
7. **Load canonical domain tables.** Preserve event timestamps and historical effective dates. Never overwrite history simply because the current source row changed.
8. **Reconcile.** Compare source totals/counts against canonical totals at meaningful control points before publishing the data.
9. **Run data contracts.** Validate keys, cardinality, referential integrity, completeness, freshness and semantic versions. Mark ACTIVE/DEGRADED/QUARANTINED.
10. **Publish analytical frames and graph projections.** Only after canonical/data-contract checks pass.
11. **Dual-run.** For critical reports, run legacy and new-platform outputs in parallel and explain material differences before cutover.
12. **Cut over incrementally.** Migrate ministry/domain slices rather than performing one irreversible government-wide big-bang move.

I.17 CDC, Refresh and History Strategy

Source type	Recommended ingestion	Canonical behavior
High-volume transactions	CDC/event stream or frequent incremental extracts.	Append immutable transaction/event facts; apply dedup/idempotency key.
Master/reference tables	CDC or periodic snapshot with hash comparison.	Create effective-dated history; do not overwrite prior business versions.
Periodic ministry reports	Batch file/API ingestion.	Store report version and reporting period; reconcile to source totals.

Source type	Recommended ingestion	Canonical behavior
Large documents/ media	Object storage with metadata record.	Warehouse stores controlled URI/object ID, classification, hash and provenance, not duplicate BLOBs in fact tables.

I.18 Performance and Physical Design

- **Use columnar storage for large analytical facts.** Government-wide transaction, encounter, payment and ridership tables will outgrow a row-store-only design.
- **Partition primarily by event/posting/fiscal date** and, where the engine supports it, cluster/sort by common entity keys such as `facility_sk`, `organization_sk`, `department_sk` or `contract_uid`.
- **Keep dimensions relatively small and reusable.** Facts contain surrogate keys; descriptive attributes live in dimensions/history tables.
- **Precompute common typed analytical frames/materialized views.** Do not make every LLM query rebuild multi-billion-row joins.
- **Separate workloads.** Ingestion, exploratory analysis, governed analytics and graph projection builds should not compete for one undifferentiated compute queue.
- **Incremental graph builds.** Update graph projections from changed canonical/bridge rows instead of rebuilding the entire graph for each ingestion.
- **Statistics and query plans.** Maintain optimizer statistics and workload-specific clustering/indexes based on actual query telemetry.
- **Do not index everything.** Excessive secondary indexes make ingestion and CDC expensive; design according to the physical engine and workload.

I.19 Security, Privacy and Purpose Enforcement

Control	Required posture
Encryption	Encrypt data in transit and at rest; use stronger key separation for <code>secure_identity</code> and law-enforcement enclaves.
Column/row security	Mask or deny sensitive identifiers/records based on actor, purpose, ministry scope and consequence class.
Purpose-based access	Access is not granted merely because the data exists in the central platform. Graph edge types and cross-domain joins require purpose authorization.
Least privilege	LLM execution identities are read-only and receive only approved views/primitives.
Audit	Log data accessed, graph projection, query plan, user/purpose, method, result reference and export/download events.
PII minimization	Use person tokens, account tokens and controlled address/location representations when full identifiers are unnecessary.
Retention/deletion	Retention follows source/legal policy; data lineage must allow affected derived artifacts to be found when records are corrected or lawfully removed.

I.20 How the LLM Uses This Database Design

```

USER QUESTION
"Which hospitals received more operating funding without comparable output growth?"
    |
    v
LLM -> SEMANTIC K-GRAPH
discovers:
Hospital -> analytics.health_hospital_funding_year
Hospital -> analytics.health_hospital_output_year
Hospital -> health.facility_dim
Hospital -> District through approved facility-area history/bridge
FiscalYear -> core.fiscal_period
    |
    v
METHOD ROUTER + TYPE/METHOD CHECKS
selects governed hospital funding/output method
    |
    v
EXECUTION PLAN
funding frame: Hospital x FiscalYear
output frame: Hospital x FiscalYear x ServiceType
registered output aggregation -> Hospital x FiscalYear
equal-grain join -> peer/trend method
    |
    v
SQL / OLAP EXECUTION ON ANALYTICS LAYER
    |
    v
RESULT + PROVENANCE + DATA-CONTRACT VERSIONS
    |
    v
LLM ANALYSIS / EXPLANATION

```

For a procurement network question, the flow changes slightly: the operational graph first returns candidate vendor/entity IDs from **graph.edge_projection**; those IDs are then applied as an entity-set filter to procurement/finance analytical facts. The graph finds relationships; the warehouse computes monetary, temporal and peer-based outcomes.

I.21 Recommended Query Surfaces

Consumer	Permitted database surface
Governed LLM lane	Registered analytics.* frames/primitives + approved core/reference views + registered graph projections.
Exploratory analyst/LLM lane	Read-only semantic views over canonical domain schemas with strict resource/row/export limits; results remain exploratory.
Data engineers	Raw/staging/canonical according to role; write through controlled pipelines only.
Graph builders	Canonical core/domain/bridge inputs and graph projection output tables.
Identity stewards	secure_identity and match audit according to dedicated authorization.
Auditors	audit/meta lineage and reproducibility records; sensitive payload only where separately authorized.

I.22 Database Anti-Patterns to Reject

- One flat "government_data" table containing mixed ministries and mixed grains.
- One generic entity-attribute-value schema used for operational analytics.
- Using PAN/Aadhaar/phone/account/source IDs directly as cross-government primary keys.
- Overwriting district, department, school type, referral level or organizational assignment history.
- Letting graph edges become the authoritative copy of business facts.
- Putting ratios/percentages in facts without retaining governed numerator/denominator where recomputation is required.
- Mixing victim/witness/suspect/accused or other materially different roles into a generic "person associated with case" status.
- Allowing every ministry to redefine District, FiscalYear, Organization or Program independently.
- Giving the LLM unrestricted credentials against raw or secure identity schemas.
- Migrating low-quality source data and then treating successful central ingestion as data certification.
- Building every possible cross-domain relationship in advance; publish only policy-authorized, typed observations and graph projections.

I.23 Initial Implementation Slice

For the first production pilot, implement the shared spine once and onboard two domains deeply rather than shallowly onboarding every ministry.

Foundation	Procurement/Finance pilot	Health pilot
meta.source_system + source_record_xref core.ministry/department/administrative_area/ organization/fiscal_period secure_identity token service data-contract registry graph projection tables audit/execution lineage	contract, bid, award, amendment, payment, budget/expenditure facts vendor organization role account/director/address observations where authorized vendor award/payment analytical frames	facility, facility-area history, funding, service activity, capacity/staffing facts hospital funding/output analytical frames district and fiscal conformance

Acceptance criterion

A ministry is not considered "migrated" when its files have merely been copied centrally. It is migrated when source reconciliation passes, canonical grain is documented, identity/reference mappings are governed, required data contracts are healthy, lineage is available, and the approved semantic/analytical/graph surfaces can reproduce agreed reference outputs.

I.24 Final Database Design Rule

Architecture rule

Centralize storage and governance, not semantic ambiguity. Preserve each domain's natural business grain; use shared identities/reference dimensions only where meaning is truly common; make cross-domain relationships explicit, temporal and provenance-bearing; publish governed analytical frames for the LLM; and derive K-Graph projections from canonical data instead of using the graph as the database of record.